# CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS

## IN EUROPEAN BANKING

**Research Paper**


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|---|---|
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| **Dataset** | European Bank Customer Dataset (10,000 records) |
| **Tools** | Python, Pandas, Scikit-learn, Matplotlib, Streamlit |
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# Abstract

Customer churn is one of the most significant threats to revenue stability in retail banking. This paper presents a comprehensive, segmentation-driven analysis of customer churn across three European geographies — France, Germany, and Spain — using a dataset of 10,000 bank customers. Through systematic data validation, feature engineering, exploratory data analysis (EDA), KPI computation, and machine learning modelling, we identify the key drivers and high-risk customer segments that contribute disproportionately to churn. Our findings reveal that Germany carries the highest geographic churn risk, the 46–60 age cohort exhibits the sharpest churn tendency, high-balance customers churn at a significantly elevated rate, and member inactivity is the strongest single-variable predictor of churn. A Random Forest classifier achieves the best predictive performance, with actionable retention strategies proposed for each high-risk segment.


# 1. Introduction

## 1.1 Background

Customer retention is fundamentally more cost-effective than customer acquisition. Industry studies consistently show that acquiring a new bank customer costs five to seven times more than retaining an existing one. Despite this, many banks manage churn reactively — responding only after a customer has already left — rather than proactively identifying at-risk customers and intervening in advance.

European retail banks face a particularly complex churn landscape. Regulatory diversity, geographic competition, digital banking disruption, and varying customer expectations across France, Germany, and Spain create a multi-dimensional churn problem that generic, one-size-fits-all retention strategies cannot adequately address.

## 1.2 Problem Statement

This project addresses the following core questions:

- What is the overall rate of customer churn across the European bank portfolio?
- Which customer segments — defined by geography, age, credit profile, tenure, and balance — exhibit the highest churn rates?
- How do churned customers differ in financial and behavioural profile from retained customers?
- Which segments represent the highest revenue risk due to churn?
- Can machine learning accurately predict which customers are likely to churn?

## 1.3 Objectives

**Primary Objectives**

- Measure the overall and segment-wise churn rate
- Identify churn distribution across all defined segmentation dimensions
- Compare churn behaviour across European regions

**Secondary Objectives**

- Understand churn concentration among high-value customers
- Evaluate the relationship between engagement, tenure, and churn
- Build predictive models to support proactive retention strategy

# 2. Dataset Description

The dataset contains **10,000 customer records** from a European retail bank with 13 original analytical columns:

| Column | Type | Description |
|---|---|---|
| `CustomerId` | Integer | Unique identifier (excluded from predictive modelling) |
| `Surname` | String | Customer surname (excluded — non-analytical) |
| `CreditScore` | Integer | Customer credit score (350–850) |
| `Geography` | Categorical | Country: France, Germany, Spain |
| `Gender` | Categorical | Male / Female |
| `Age` | Integer | Customer age in years |
| `Tenure` | Integer | Years as an account holder (0–10) |
| `Balance` | Float | Current account balance |
| `NumOfProducts` | Integer | Number of bank products held (1–4) |
| `HasCrCard` | Binary | Owns a credit card (0/1) |
| `IsActiveMember` | Binary | Active member indicator (0/1) |
| `EstimatedSalary` | Float | Estimated annual salary |
| `Exited` | Binary | Churn indicator — **target variable** (0 = retained, 1 = churned) |

**Data Quality Assessment**

- No missing values detected across all columns
- Zero duplicate `CustomerId` records confirmed
- All binary variables (`HasCrCard`, `IsActiveMember`, `Exited`) verified as valid 0/1 integers
- Class distribution: 7,963 retained (79.63%), 2,037 churned (20.37%)

# 3. Methodology

## 3.1 Data Ingestion & Validation

The dataset was loaded into a Pandas DataFrame from `European_Bank.csv`. Null checks, duplicate detection, and category validations were executed to ensure dataset integrity.

## 3.2 Feature Engineering — Segmentation Dimensions

Four derived categorical segmentation columns were engineered:

| New Column | Source | Segments & Cut-offs |
|---|---|---|
| `AgeGroup` | `Age` | `<30`, `30–45`, `46–60`, `60+` |
| `CreditScoreBand` | `CreditScore` | Low (<580), Medium (580–669), High (≥670) |
| `TenureGroup` | `Tenure` | New (0–2 yrs), Mid-term (3–6 yrs), Long-term (7+ yrs) |
| `BalanceSegment` | `Balance` | Zero-balance (=0), Low-balance (>0 ≤ median), High-balance (>median) |

The positive-balance median was calculated exclusively among accounts with balances greater than zero (~€119,839.69), ensuring zero-balance accounts did not skew the threshold.

### 3.3 Key Performance Indicator (KPI) Definitions

| KPI | Formula / Operational Definition |
|---|---|
| **Overall Churn Rate** | `(Total Churned / Total Customers) × 100` |
| **Segment Churn Rate** | Churn rate calculated within each distinct subgroup |
| **High-Value Churn Ratio** | Churn percentage among customers in the `High-balance` segment |
| **Geographic Risk Index** | `Country Churn Rate / Overall Churn Rate` (>1 indicates above-average risk) |
| **Engagement Drop Indicator** | `Inactive Member Churn Rate – Active Member Churn Rate` (percentage points) |

### 3.4 Predictive Modeling Architecture

An 80/20 stratified split was created to preserve target class proportions:

- **Baseline Model:** Logistic Regression with `class_weight='balanced'` and `StandardScaler` numeric scaling.
- **Ensemble Model:** Random Forest Classifier (300 estimators, `class_weight='balanced'`).
- **Evaluation Criteria:** Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

## 4. Key Findings & KPI Results

### 4.1 Overall Churn Benchmark

- Total Customers: **10,000**
- Total Churned: **2,037**
- **Overall Churn Rate: 20.37%** (approximately 1 in every 5 customers leaves the institution).

### 4.2 Geographic Segmentation & Risk Index

| Geography | Total Customers | Churned | Churn Rate | Geographic Risk Index |
|---|---|---|---|---|
| **Germany** | 2,509 | 814 | **32.44%** | **1.59 (High Risk)** |
| **Spain** | 2,477 | 413 | **16.67%** | **0.82 (Low Risk)** |
| **France** | 5,014 | 810 | **16.15%** | **0.79 (Low Risk)** |

> ***Key Insight:*** *Customers in Germany churn at double the frequency of customers in France and Spain. Germany accounts for 39.96% of all churned accounts despite representing only 25.09% of the customer base.*

### 4.3 Age Group Disparity

| Age Group | Customers | Churned | Churn Rate |
|---|---|---|---|
| `<30` | 1,668 | 126 | **7.55%** |
| `30–45` | 6,367 | 884 | **13.88%** |
| `46–60` | 2,234 | 967 | **43.29%** |

| Age Group | Customers | Churned | Churn Rate |
|---|---|---|---|
| 60+ | 731 | 186 | **25.44%** |

> ***Key Insight:*** *Churn accelerates steeply past age 45. The 46–60 cohort experiences a staggering 43.29% churn rate.*

### 4.4 Financial Profile & High-Value Customer Risk

| Balance Segment | Customers | Churned | Churn Rate |
|---|---|---|---|
| Zero-balance | 3,617 | 500 | 13.82% |
| Low-balance | 3,191 | 749 | 23.47% |
| High-balance | 3,192 | 788 | **24.69%** |

- **High-Value Churn Ratio: 24.69%**
- Mean balance of churned customers: **€91,108.54**
- Mean balance of retained customers: **€72,745.30**

> ***Key Insight:*** *Retained customers carry lower average balances than churners. Customers leaving the bank are higher-wealth clients, representing major balance sheet and capital liquidity risk.*

### 4.5 Engagement & Activity Analysis

| Member Activity | Customers | Churned | Churn Rate |
|---|---|---|---|
| Inactive Member (0) | 4,849 | 1,302 | **26.85%** |
| Active Member (1) | 5,151 | 735 | **14.27%** |

- **Engagement Drop Indicator: 12.58 percentage points** (Inactive churn is almost double active churn).

### 4.6 Gender Differences

- **Female Churn Rate: 25.07%** (1,139 / 4,543)
- **Male Churn Rate: 16.46%** (898 / 5,457)
- Female customers exhibit an 8.61 percentage point higher propensity to exit.

## 5. Machine Learning Evaluation

| Metric | Logistic Regression | Random Forest Classifier |
|---|---|---|
| **Accuracy** | 70.85% | **85.90%** |
| **Precision** | 38.62% | **76.24%** |
| **Recall** | **71.25%** | 47.91% |
| **F1-Score** | 50.09% | **58.85%** |
| **ROC-AUC** | 0.771 | **0.854** |

- **Model Choice:** The **Random Forest Classifier** is selected as the primary production model due to its high precision (76.24%) and superior area under the ROC curve (0.854), minimizing costly false-positive retention offers.
- **Top Feature Drivers:** `Age`, `NumOfProducts`, `IsActiveMember`, `Balance`, and `Geography_Germany`.

## 6. Strategic Recommendations

- **German Regional Intervention Plan:**

Conduct targeted customer exit research specifically in Germany to determine if fees, digital product performance, or local fintech competitors (e.g., N26) are driving departures.

- **Wealth Management & Premier Retention (Ages 45–60):**

Deploy dedicated relationship managers and attractive term-deposit yields to accounts over €100,000 held by mature clients.

- **Automated Inactivity Triggers:**

Establish CRM alerts that flag accounts displaying no transaction activity over 60 days, sending proactive re-engagement incentives before churn takes place.

- **Product Diversification Strategy:**

Customers with a single product churn at significantly higher rates; incentivize holding 2 complementary products (e.g., combining checking with an investment/savings wallet).

## 7. Conclusion

Customer churn in retail banking is predictable through behavioral and demographic segmentation. By focusing intervention resources on German accounts, mature clients aged 46–60, and inactive high-balance savers, the bank can optimize retention spend and safeguard millions of euros in customer lifetime value.

## References and source note

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